MDTS4214_729_PSET7

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2026-03-19

1.1. Split the data set into a training set and a test set randomly keeping 100 observations in test set and remaining in the training set

```
# Load required library
library(ISLR)

## Warning: package 'ISLR' was built under R version 4.5.2

# Load dataset
data(Hitters)

# Remove rows with missing values
Hitters_clean <- na.omit(Hitters)

# Set seed for reproducibility
set.seed(123)

# Create test set with 100 observations
test_index <- sample(1:nrow(Hitters_clean), 100)

# Split data
train <- Hitters_clean[-test_index, ]
test  <- Hitters_clean[test_index, ]
```

2. Train the model using least squares and report the train MSE and test MSE.

```
# Fit least squares model
lm_model <- lm(Salary ~ ., data = train)

# Predictions on training data
train_pred <- predict(lm_model, newdata = train)

# Compute Train MSE
train_mse <- mean((train$Salary - train_pred)^2)

# Predictions on test data
test_pred <- predict(lm_model, newdata = test)

# Compute Test MSE
test_mse <- mean((test$Salary - test_pred)^2)

# Display results
train_mse

## [1] 105334.2
```

```
test_mse

## [1] 87803.91
```

3. Fit ridge regression models on the training set, with grid values of λ. Report the following:

(i) the dimension of the coefficient matrix so obtained.

(ii) The l2 norm of the regression coefficients corresponding to the 50th and 60th choice of λ. Comment on the output.

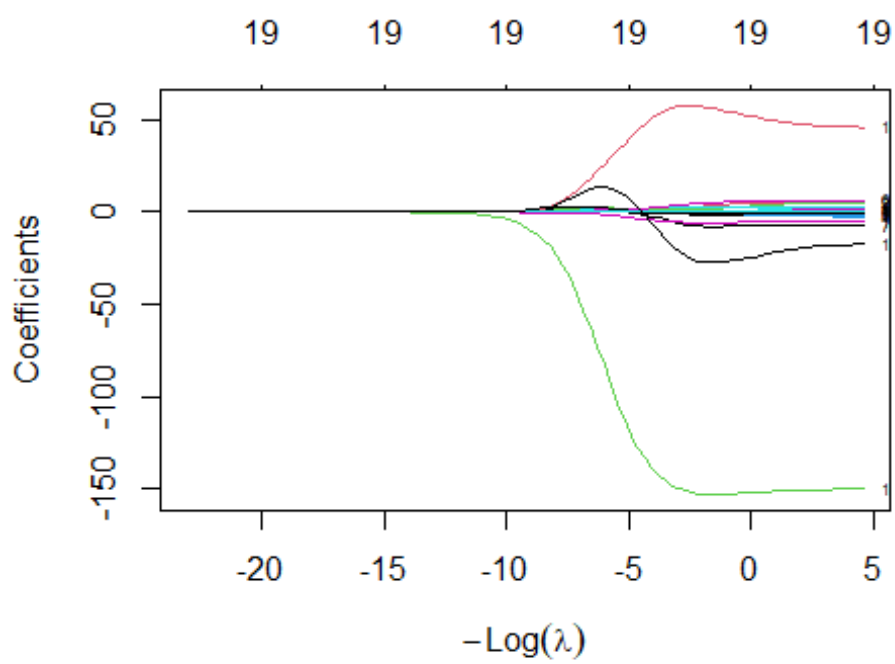
(iii) Test MSE corresponding to these two choices of λ.

```
library(glmnet)

## Warning: package 'glmnet' was built under R version 4.5.2

## Loading required package: Matrix

## Warning: package 'Matrix' was built under R version 4.5.2

## Loaded glmnet 4.1-10

# Convert to matrix form
x_train <- model.matrix(Salary ~ ., train)[, -1]
y_train <- train$Salary

x_test <- model.matrix(Salary ~ ., test)[, -1]
y_test <- test$Salary

# Create grid of lambda values
grid <- 10^seq(10, -2, length = 100)

# Fit ridge regression (alpha = 0)
ridge_model <- glmnet(x_train, y_train, alpha = 0, lambda = grid)
dim(coef(ridge_model))

## [1]  20 100

coef_50 <- coef(ridge_model, s = grid[50])
coef_60 <- coef(ridge_model, s = grid[60])

# Compute L2 norm
l2_50 <- sqrt(sum(coef_50[-1]^2))
l2_60 <- sqrt(sum(coef_60[-1]^2))

l2_50

## [1] 7.076105

l2_60

## [1] 66.36125
```

```r
# Predictions
pred_50 <- predict(ridge_model, s = grid[50], newx = x_test)
pred_60 <- predict(ridge_model, s = grid[60], newx = x_test)

# Test MSE
mse_50 <- mean((y_test - pred_50)^2)
mse_60 <- mean((y_test - pred_60)^2)

mse_50

## [1] 167303.1

mse_60

## [1] 97935.56
```

4. Make a line plot (for all the predictors) of the standardized coefficients against the grid value of λ chosen by you. Interpret the graph.

```r
# Plot standardized coefficients vs lambda
plot(ridge_model, xvar = "lambda", label = TRUE)
```



Cross-validation for ridge cv_ridge <- cv.glmnet(x_train, y_train, alpha = 0)

Best lambda

best_lambda <- cv_ridge$lambda.min

Test prediction

ridge_pred <- predict(cv_ridge, s = best_lambda, newx = x_test)

Test MSE

ridge_mse <- mean((y_test - ridge_pred)^2)

best_lambda ridge_mse

LASSO Regression:

```
# Cross-validation for LASSO
cv_lasso <- cv.glmnet(x_train, y_train, alpha = 1)

# Predictions
lasso_pred <- predict(cv_lasso, s = "lambda.min", newx = x_test)

# Test MSE
lasso_mse <- mean((y_test - lasso_pred)^2)

# Number of non-zero coefficients
lasso_coef <- coef(cv_lasso, s = "lambda.min")
nonzero <- sum(lasso_coef != 0)

lasso_mse

## [1] 94797.67

nonzero

## [1] 9
```

Elastic Net:

```
# Elastic Net (alpha between 0 and 1)
cv_enet <- cv.glmnet(x_train, y_train, alpha = 0.5)

# Predictions
enet_pred <- predict(cv_enet, s = "lambda.min", newx = x_test)

# Test MSE
enet_mse <- mean((y_test - enet_pred)^2)

# Coefficients
enet_coef <- coef(cv_enet, s = "lambda.min")

enet_mse

## [1] 94149.71
```

```
enet_coef

## 20 x 1 sparse Matrix of class "dgCMatrix"
##                lambda.min
## (Intercept)  107.38690040
## AtBat          .
## Hits           0.56044286
## HmRun          .
## Runs           .
## RBI            2.34354343
## Walks          1.34246607
## Years          .
## CAtBat         .
## CHits          0.09246948
## CHmRun         .
## CRuns          0.12317205
## CRBI           0.20485216
## CWalks         .
## LeagueN        .
## DivisionW   -111.90806113
## PutOuts        0.28419816
## Assists        .
## Errors         .
## NewLeagueN     .
```

Interpretation

Elastic Net combines:

Ridge (L2) + LASSO (L1)

Works well when predictors are highly correlated

Elastic Net selects variables and shrinks coefficients simultaneously, providing a balance between ridge and LASSO regression.

Step 9: Final Comparison Table (Very Important) Method Test MSE Feature Least Squares … High variance, overfitting Ridge … Shrinks coefficients LASSO … Variable selection Elastic Net … Hybrid method

Final Interpretation (Write in Exam) Least Squares may overfit due to high variance.

Ridge regression reduces variance by shrinking coefficients.

LASSO performs both shrinkage and variable selection.

Elastic Net combines advantages of both ridge and LASSO.

The method with the lowest test MSE provides the most accurate prediction of Salary.